

RiskIntel AI

From financial text to portfolio stress

A financial-text risk engine linked to a scenario-based, asset-level stress model.

CANDIDATE

Aditya Saurav

Birla Institute of Technology, Mesra

btech60105.23@bitmesra.ac.in

PROTOTYPE AT A GLANCE

Financial news +
financial-social text

RISK CLASSIFICATION

9 event types

SAVED STRESS EXAMPLE

\$10.0m

6 synthetic holdings

RiskIntel_AI repository

A risk headline does not show portfolio exposure

The missing step is a consistent way to turn an event into position-level stress.

BUSINESS PROBLEM

What does this news mean for the book?

Unstructured input

News arrives as headlines and financial commentary.

Decision gap

Sentiment alone does not estimate holding-level loss.

Repeatability

Manual scenario assignment is difficult to compare.

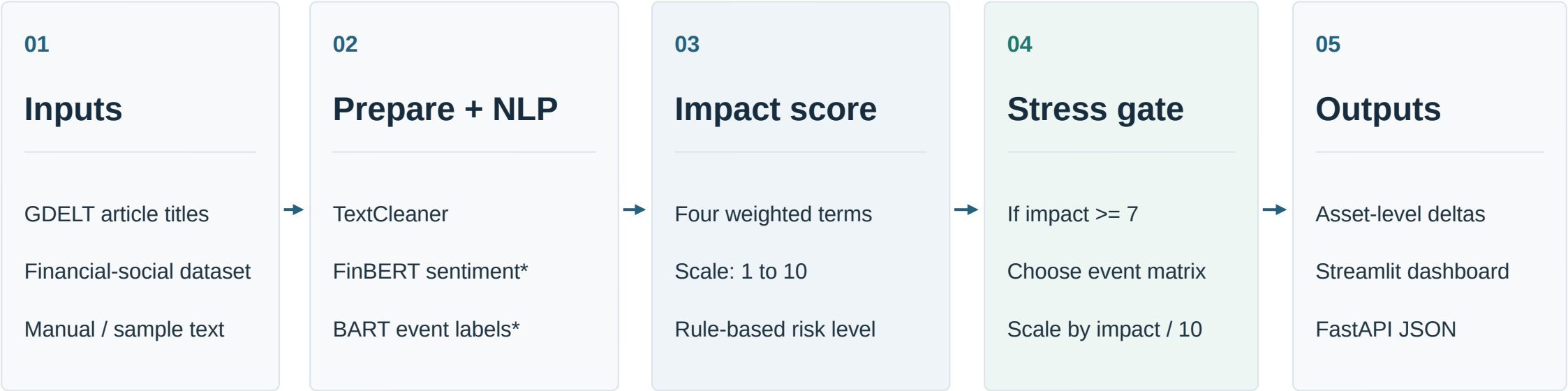
RISKINTEL WORKFLOW

- | | | |
|----|------------|--|
| 01 | Ingest | GDELT titles, sample data or manual text |
| 02 | Interpret | Sentiment and nine event categories |
| 03 | Prioritize | Weighted impact score between 1 and 10 |
| 04 | Stress | At score 7 or above, revalue a portfolio |

Result: a structured risk signal, followed by a stress estimate when the trigger is met.

From source text to a holding-level stress estimate

The pipeline uses separate data, signal and stress-result structures.



INTERMEDIATE DATA

DataRecord

Source adapters



RiskSignal

Event, sentiment, impact



StressTestResult

Portfolio and asset-level changes

* Model inference uses rule-based fallbacks on failure; manual input does not require DataRecord.

Implementation choices, with their trade-offs

Models support interpretation; explicit Python rules determine how stress is applied.

01 INPUT + PREPARATION

GDELT returns indexed news titles.

A financial-social dataset supplies text posts.

Regex, entity mapping and optional MD5 batch dedup.

02 FINANCIAL NLP

ProsusAI/FinBERT: signed sentiment estimate.

BART-large-MNLI: nine financial event labels.

Both modules fall back to keyword rules on error.

03 EXPLAINABLE PRIORITY

Four weighted inputs; score constrained to 1-10.

Severity 35% + sentiment magnitude 20%.

Entity importance 25% + urgency 20%.

04 SIMULATOR + DELIVERY

Event-specific shock matrices; impact / 10 scaling.

Six-asset scenario with a per-holding breakdown.

Streamlit, Plotly, FastAPI and CSV / JSON exports.

Impact score = 0.35 x severity + 0.20 x |sentiment| x 10 + 0.25 x entity + 0.20 x urgency

Sentiment magnitude is capped at 10; score is bounded to [1, 10]. The shock rates are scenario inputs, not fitted forecasts.

Saved scenario: a \$454,250 modeled loss

Geopolitical shock; impact input 7.9 / 10; six-holding synthetic portfolio.

PORTFOLIO BEFORE		PORTFOLIO AFTER	NET CHANGE
\$10,000,000		\$9,545,750	-4.5425%
SAVED ASSET-LEVEL CHANGES		USD	CHECKED-IN OUTPUT
Equity portfolio		-\$158,000	<pre>"event_type": "Geopolitical" "impact_score": 7.9 "triggered": true "total_loss": 454250.0</pre>
Corporate bond		-\$94,800	
Derivative A		-\$94,800	
Corporate loan A		-\$79,000	
Corporate loan B		-\$59,250	
Government bond		+\$31,600	
WORKFLOW COMPARISON		Manual: label, choose shocks, calculate. Prototype: structured signal and revaluation. Speed advantage has not been measured.	

Source: saved stress_test_result.json; results are hypothetical, not observed portfolio performance.

A clearer starting point for a risk review

The output connects an event category to named portfolio positions and dollar changes.

REVIEW PACKET: GEOPOLITICAL SCENARIO

What deserves attention?

Trigger	Impact input 7.9; threshold 7.0
Largest loss	Equities: -\$158,000
Partial offset	Government bond: +\$31,600
Net effect	-\$454,250 across six holdings

POTENTIAL USERS

Risk analyst

Sort events by severity and inspect the scoring components.

Portfolio manager

See modeled exposure by holding before considering mitigation.

Review and engineering teams

Inspect JSON fields and scenario parameters for further testing.

This prototype supports human review; it does not select trades or measure avoided losses.

What is established, and what needs validation

The saved stress arithmetic checks out; forecasting accuracy has not been tested.

CURRENT LIMITATION	NEXT STEP
GDELT provides titles; social text comes from a dataset rather than a live X feed.	Capture original timestamps, links and source provenance during ingestion.
Model failures invoke keyword rules; the repo includes no label or latency benchmark.	Evaluate predictions and runtime in both model and fallback modes.
Scenario shocks are predefined and scale linearly with the impact input.	Backtest scenarios against historical returns and quantify model error.
The urgency rule checks singular keywords; the plural word tariffs is not a match.	Handle word variants consistently and add tests for singular and plural cases.
Both \$10m and \$3.5m portfolio fixtures exist; the selected default depends on code path.	Use one explicit portfolio configuration across the demo, dashboard and API.
Current scope: analyst-reviewed stress testing, not a trading system or validated loss forecast.	